**MERN STACK**

# TRAINING TR-103 REPORT DAY 17

**14 July 2026**

## Overview:

The seventeenth day of the MERN Stack training focused on React Router and the `useState` Hook. We learned how to implement client-side routing using React Router, allowing users to navigate between different pages without reloading the application. The session also introduced the `useState` Hook, which enables functional components to manage dynamic data and update the user interface efficiently. Practical exercises helped us build multi-page React applications with interactive components.

## Learning Objectives:

- Understand the concept of React Router.
- Learn the purpose of `BrowserRouter`, `Routes`, and `Route`.
- Create navigation using the `Link` component.
- Understand the `useState` Hook.
- Learn how to manage state in React functional components.

## REACT

### React Router

React Router is a library used for client-side routing in React applications. It allows users to navigate between different pages without refreshing the entire webpage.

Features:

- Enables client-side navigation.
- Improves user experience.
- Supports dynamic routing.

## BrowserRouter

BrowserRouter is the top-level component that enables routing using the browser's History API. It wraps the entire React application so that routing can be managed.

Features:

- Enables routing in React.
- Uses the browser's history.
- Wraps the main application.

Syntax:

```
import { BrowserRouter } from "react-router-dom";
function App() {
 return (
  <BrowserRouter>
        {/* All routes go here */}
  </BrowserRouter>
);
}
```

## Routes

The Routes component acts as a container for all Route components. It displays the component that matches the current URL.

Features:

- Holds multiple routes.
- Renders the matching route.
- Organizes application routing.

Syntax:

```
<Routes>
    <Route path="/" element={<Home />} />
    <Route path="/about" element={<About />} />
</Routes>
```

## Route

A Route maps a URL path to a React component.

Important Properties:

- path – Specifies the URL path.
- element – Specifies the component to display.

Syntax:

```
<Route path="/about" element={<About />} />
```

## Link

The Link component is used to navigate between pages without reloading the webpage.

Features:

- Creates navigation links.
- Updates the URL without refreshing.
- Improves navigation in React applications.

Syntax:

```
<Link to="/about">About</Link>
```

## useState Hook

The useState Hook allows functional components to store and update data that can change over time.

Features:

- Manages component state.
- Updates the user interface dynamically.
- Simplifies state management in functional components.

Components of useState:

- State Variable – Stores the current value.
- State Function – Updates the stored value.
- Initial Value – Starting value of the state.

Syntax:

```
import { useState } from "react";
```

# TASK :

```jsx
import { useState} from "react"

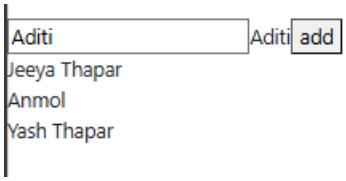
export const State = () =>{
    const[name,setname] = useState("Jeeya Thapar")
    const[na,setna] = useState([])

    const ab=()=>{
        setna([...na,name])

    }

    return(
        <>
        <h1>Name: {name}</h1>
        <input type ="text" value ={name} onChange={(e)=>setname(e.target.value)}/>{name}
        <button onClick={ab}>add</button>
        {
            na.map((a,index) =>
                <li key={index}>{a}</li>
                )
        }
        </>
    )
}
```

**Output :**




# Conclusion:

By the end of Day 17, I gained a clear understanding of React Router and the `useState` Hook. I learned how to build multi-page React applications using client-side routing and manage dynamic data within functional components, making React applications more interactive and user-friendly.